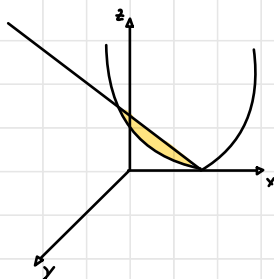


Esercizio I: [6 pt]

Calcolare il volume del solido

$$S = \{(x, y, z) \in \mathbb{R}^3, (x-3)^2 + y^2 \leq z \leq 18-6x\}.$$



$$V = \iiint 1 \, dx \, dy \, dz$$

$$\text{con } (x-3)^2 + y^2 \leq z \leq 18-6x$$

Adesso va stabilito E

$$x^2 - 6x + 9 + y^2 < 18 - 6x$$

$$x^2 + y^2 - 9 = 0$$

cambio in c/o polari

$$0 < \rho^2 < 9$$

$$0 < \rho < 3 \quad \text{mentre} \quad 0 < \theta < 2\pi$$

$$\begin{cases} 0 < \rho < 3 \\ 0 < \theta < 2\pi \end{cases}$$

$$dx \, dy = \rho \, d\rho \, d\theta$$

$$\int_0^3 \int_0^{2\pi} \left(\int_{(x-3)^2 + y^2}^{18-6x} dz \right) \rho \, d\rho \, d\theta$$

$$\int_0^3 \int_0^{2\pi} [18-6x - (x-3)^2 - y^2] \rho \, d\rho \, d\theta$$

$$\int_0^3 \int_0^{2\pi} [18-6x - x^2 + 6x - 9 - y^2] \rho \, d\rho \, d\theta$$

$$- \int_0^3 \int_0^{2\pi} [x^2 + y^2 - 9] \rho \, d\rho \, d\theta$$

$$- \int_0^3 \int_0^{2\pi} \rho^3 - 9\rho \, d\rho \, d\theta = -2\pi \left[\frac{1}{4} \rho^4 - \frac{9}{2} \rho^2 \right]$$

$$= -2\pi \left[\frac{81}{4} - \frac{81}{2} \right] = \frac{81}{2} 2\pi = \frac{81}{2} \pi \sqrt{}$$

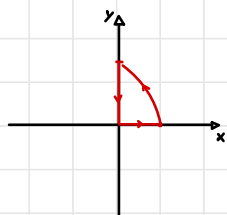
Esercizio II: (pt 10)

Calcolare l'area della regione del piano compresa tra gli assi cartesiani ed il grafico della curva parametrizzata da $(x(t), y(t)) = (\cos t, t(\sin t)^2)$, $t \in [0, \frac{\pi}{2}]$.

$$x(t), y(t) = [\cos t, t(\sin t)^2] \quad \text{con} \quad t \in [0, \frac{\pi}{2}]$$

$t=0$ sono in $1, 0$

$t = \frac{\pi}{2}$ sono in $0, \frac{\pi}{2}(1)$



quindi ho $y(t) = [\cos t, t(\sin t)^2]$

$$\dot{y}(t) = (-\sin t, (\sin t)^2 + 2t \sin t \cos t)$$

Prendo $F = (-y, 0)$ in modo che $\text{rot}(F) = 1$

adesso posso scrivere che $L(F, \gamma) = \int_{\gamma} F(\gamma) \dot{\gamma} dt = \underbrace{\iint 1 dx dy}_{\text{area}}$

quindi uso green-gours per dire che:

$$\iint 1 dx dy = \int M dx + \int N dy$$

$$= \int M(\gamma(t)) \dot{x} + \int N(\gamma(t)) \dot{y}$$

$$= \int_0^{\frac{\pi}{2}} t(\sin t)^2 (-\sin t) dt + \int 0$$

$$= \int_0^{\frac{\pi}{2}} -t(\sin^3 t) dt = \left| -\frac{t}{12} (\cos(3t) - 9\cos t) \right|_0^{\frac{\pi}{2}} + \int \sin^3 t dt$$

$\downarrow$ $\downarrow$
 f g
 $g = \int g =$

$$= |0| + \frac{1}{12} (\cos(3t) - 9)$$

Esercizio III: (pt 4)

Data la funzione

$$f(x, y) = 9x^2 + 36x^2y - 4y - 1,$$

trovare i punti critici, determinando se siano massimi o minimi relativi o selle.

$$f(x, y) = 9x^2 + 36x^2y - 4y - 1$$

$$\nabla f(x, y) = 18x + 72xy ; 36x^2 - 4$$

$$\begin{cases} 18x + 72xy = 0 \\ 36x^2 - 4 = 0 \end{cases} \quad \begin{aligned} &\text{if } x = \frac{1}{3} \quad \frac{18}{3} + \frac{72}{3}y = 0 \quad 24y = -6 \quad y = -\frac{1}{4} \quad p_1 = \left(\frac{1}{3}; -\frac{1}{4}\right) \\ &\text{if } x = -\frac{1}{3} \quad -6 - 24y = 0 \quad y = -\frac{1}{4} \\ &x = \pm \frac{1}{3} \end{aligned}$$

$$\begin{aligned} p_1 &= \left(\frac{1}{3}; -\frac{1}{4}\right) \\ p_2 &= \left(-\frac{1}{3}; -\frac{1}{4}\right) \end{aligned}$$

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}$$

con

$$\frac{\partial f}{\partial x} = 18x + 72xy$$

$$\frac{\partial^2 f}{\partial x^2} = 18 + 72y$$

$$\frac{\partial^2 f}{\partial x \partial y} = 72x$$

$$\frac{\partial f}{\partial y} = 36x^2 - 4$$

$$\frac{\partial^2 f}{\partial y^2} = 0$$

$$\frac{\partial^2 f}{\partial y \partial x} = 72x$$

$$H = \begin{bmatrix} 18 + 72y & 72x \\ 72x & 0 \end{bmatrix}$$

$$H(p_1) = \begin{bmatrix} 0 & 24 \\ 24 & 0 \end{bmatrix}$$

selle per $\det < 0$

$$H(p_2) = \begin{bmatrix} 0 & -24 \\ -24 & 0 \end{bmatrix}$$

selle

Esercizio I: [4 pt]

Si consideri la funzione

$$f(x, y) = \begin{cases} \frac{x^{\frac{1}{3}} y^2}{x^2 + y^6} \sin(x) & \text{se } (x, y) \neq (0, 0) \\ 0 & \text{se } (x, y) = (0, 0) \end{cases}$$

Stabilire se la funzione è differenziabile in $(0, 0)$.

a) Continuità

$$y^3 = x$$

$$\lim_{y \rightarrow 0} \frac{y^{\cancel{3}}}{2y^{\cancel{6}}} = \frac{1}{2}$$

Non è diff perché non è continua

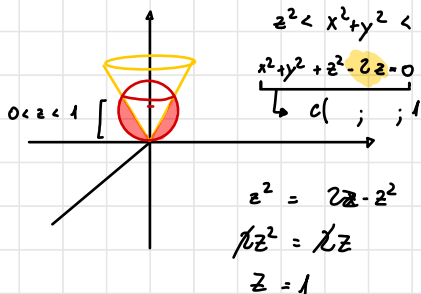
Esercizio II: (pt 8)

Calcolare

$$\iiint_A \frac{1}{\sqrt{x^2 + y^2 + z^2}} dx dy dz$$

dove $A = \{(x, y, z) \in \mathbb{R}^3, z^2 \leq x^2 + y^2 \leq 2z - z^2\}$.

$$\begin{cases} x = p \cos \theta \\ y = p \sin \theta \\ z = z \end{cases}$$



$$z^2 \leq x^2 + y^2 <$$

$$c = \left(-\frac{a}{2}\right)$$

$$x^2 + y^2 + z^2 - 2z = 0$$

$$c = 0, 0, 1$$

$$r = \sqrt{1^2} = 1$$

$$z^2 = 2z - z^2$$

$$4z^2 = 2z$$

$$z = 1$$

$$z < p < \sqrt{2z - z^2}$$

$$0 < \theta < 2\pi$$

$$\iiint_A \frac{1}{\sqrt{p^2 + z^2}} p dp d\theta dz$$

$$2p(p^2 + z^2)^{-\frac{1}{2}}$$

$$(p^2 + z^2)^{\frac{1}{2}}$$

$$\int_0^1 \int_0^{2\pi} \int_z^{\sqrt{2z-z^2}} \frac{1}{\sqrt{p^2 + z^2}} p dp d\theta dz$$

$$= \frac{1}{2} \int_0^1 \int_0^{2\pi} \frac{(p^2 + z^2)^{\frac{1}{2}}}{\sqrt{p^2 + z^2}} d\theta dz$$

$$= \int_0^1 \int_0^{2\pi} \left| (p^2 + z^2)^{\frac{1}{2}} \right|_{\sqrt{2z-z^2}}^{\sqrt{2z-z^2}} d\theta dz$$

$$= \int_0^1 \int_0^{2\pi} (2z)^{\frac{1}{2}} - (2z)^{\frac{1}{2}} d\theta dz$$

$$\frac{1}{2}$$

$$2p(p^2 + z^2)^{-\frac{1}{2}}$$